

Do Temporal Network Signals Predict How Scientific Concepts Spread Across Disciplines?

Executive Summary

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Headline Finding

Early ego-network openness ($\text{OPEN}_{\text{home}}$) predicts cross-field breadth in a descriptive evidence synthesis across six non-selection bodies: DerSimonian–Laird pooled partial Spearman $\rho = +0.069$ [0.038, 0.100], $I^2 = 0$, 6/6 bodies positive. The outcome is exploratory.

Study Design

The study used the OpenAlex bulk snapshot (476,196,327 works; 2026-09-23) to identify 12,499 scientific concepts, track 27,393 adoption episodes across 26 venue fields, and screen 53 early network indicators against a five-feature popularity baseline (B5). The primary outcome was rarefied field breadth (O_{2r} , $m = 50$). A development panel (DEV: CS, Engineering, BGM, Medicine homes; onset 2003–2009; 4,771 concepts) was used to freeze indicator rankings; four held-out field groups (Physical Sciences, Life & Environment, Social Sciences, Math & Decision; 3,372 concepts), one onset cohort (2010–2014; 4,356 concepts), and one fresh cohort (2015–2017; 1,443 concepts) served as non-selection bodies.

Iteration-by-Iteration Results

Iteration 1 screened five rival mechanisms on 46–48 DEV concepts. All five failed the pre-registered $\Delta\rho$ rule: naturalisation gap A_h^* ($\Delta\rho = -0.006$), co-occurrence diversity D_{ratio} (+0.006), selectivity F_{res} (−0.060), and gateway landing G (+0.033). The field-level gateway-retention signal ($\Delta\text{AUC} +0.103$, 80 episodes) became the lead for iteration 2. Two artifacts failed (dataset 1, experiment 2).

Iteration 2 built the authoritative 12,499-concept frame (zero API credits) and ran the decisive test. Gateway retention was **disconfirmed**: held-out $\Delta\text{AUC} = -0.00001$ [−0.0006, +0.0003] on 27,393 episodes. The iteration-1 lead could not be certified above chance (shuffled- R placebo 95th percentile exceeded the original +0.103). However, retained-field relatedness predicted the next field entered: $d_0 = 0.302$ [0.240, 0.369] on 653 concepts. External recognition (O5) was unrelated to breadth.

Iteration 3 confirmed the retained-field relatedness finding on an independent frame (11,841 concepts): pooled held-out $d_0 = 0.322$ [0.291, 0.355], $\text{LR} = 325.8$ ($p < 10^{-70}$). However, the volume-matched contrast was null (−0.028 [−0.105, +0.046]) and the coefficient reversed under

minimum-conditional-probability proximity: the result is backbone-specific. The 53-indicator held-out screen confirmed 7 of 10 DEV-frozen indicators (5 positive, 2 negative). ElasticNet added Spearman +0.059 over B5 on held-out. Experiment 9 (trajectories) failed.

Iteration 4 tested OPEN on a fresh 2015–2017 cohort: OPEN_{home} partial Spearman +0.091 [+0.013, +0.171] at R2, but the DL pool over groups was +0.083 [−0.007, +0.173], and it added no practical prediction (+0.002 over B5 Spearman of 0.768). The breadth decomposition showed early contact diversity accounted for 73% of the gap; no discrete trajectory typology survived. The within-concept closure test (Experiment 11) was not supported: density $b = -0.070$ [−0.180, +0.040]; OPEN $b = +0.015$ [−0.038, +0.069].

Iteration 5 tested OPEN on 636 vocabulary-free newborn concepts (Frame N): OPEN_{home} PSP +0.117 [+0.020, +0.218] at R3. The Cheng consistency reversal showed that ideational consistency predicts next-year volume (+53.5%/SD) but *narrower* cross-field reach (PSP −0.069 [−0.093, −0.047], 5/5 groups negative, $I^2 = 0$). The within-concept permutation null absorbed the year-to-year churn component (excess NOVCHURN PSP = +0.008 vs raw +0.116): what predicts breadth is static topical non-redundancy, not temporal partner turnover. However, V2 excess split-half reliability was ~ 0.01 – 0.05 , so this test could not adjudicate temporal churn at ~ 10 papers/year.

Key Results

Finding	Effect	95% CI	Status
OPEN _{home} → breadth (DL, $k = 6$)	+0.069	[+0.038, +0.100]	Exploratory
NOVCHURN _{home} (DL, $k = 5$)	+0.105	[+0.069, +0.140]	Exploratory
Consistency → breadth (DL, $k = 5$)	−0.069	[−0.093, −0.047]	Reversal
Retained-field relatedness d_0	+0.322	[+0.291, +0.355]	Confirmed
Gateway retention	−0.00001	[−0.0006, +0.0003]	Disconfirmed
OPEN _{home} forecasting gain	+0.002	[−0.003, +0.008]	Null

Failures and Dead Ends

- All five iteration-1 concept-level candidates (A_h^* , D , F , G , S) failed the pre-registered screen or were untested.
- Gateway centrality for field-level retention was a domain-specific proxy; disconfirmed at scale.
- External recognition (O5) was unrelated to any publication outcome.
- No discrete trajectory typology passed the naming rule (DTW vs HMM ARI 0.094–0.222).
- Within-concept closure (Exp11) was null on DEV and opposite-signed on held-out.
- 4 of 20 artifacts failed or were interrupted.
- All reviews scored 2–3 (blocking), primarily for conclusions contradicting the artifacts.

Open Questions

- **Temporal vs static:** V2 excess reliability was too low to decide whether temporal churn adds information beyond static topical dispersion.
- **Mechanism:** association, not causation, was shown. OPEN could reflect intrinsic generality.
- **Pre-onset footprint:** roughly half of the retained-frontier indicators' signals came from pre-onset field history.
- **Vocabulary bias:** Frame N was small (636 concepts, pre-unseal power 0.47).
- **Domain heterogeneity:** retained-field relatedness $I^2 = 0.92$ across groups.